



Academic Exchange

Nano-crystal glass ceramic of high performance prepared from the residual waste from mineral resources

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Unit Brief Introduction

❖ What Does KLIEBR Stand For?

KL - IE - BR

Key Laboratory

Integrated
Exploitation

Bayan-Obo
Resources

Unit Brief Introduction

KLIEBR started at the beginning of the 90's as Metallurgy Research Institute, focusing on the simulation of metallurgy process with momentum, heat, mass transmissions.

3-dimensional calculation software was developed and successfully applied in tundish, mould, LF furnace and RH furnace.

2003, Key Lab of Metallurgical Engineering ;

2004, Key Lab of Advanced Metallic Material ;

2007, Key Lab of Clean Combustion;

in 2007, KLIEBR became the key lab of Ministry of education.

Unit Brief Introduction

省部共建国家重点实验室培育基地

内蒙古自治区白云鄂博矿多金属资源综合利用
重点实验室

科学技术部



**Aug 2010,
Became Lab of Ministry of Science and Technology**

**Spring 2012,
Exclusive Building came into service**



Unit Brief Introduction

KLIEBR' s Mission: Clean, High value-added, comprehensive utilization of BR

Problems:

- ❖ Niobium, an advanced alloying element, has not been utilized
- ❖ Thorium, a new nuclear fuel, radioactive contamination, has not been utilized
- ❖ Tailings, BF slags and other solid waste take up a lot of valuable land resources

Products and Performance



The pilot plant of slag glass-ceramics

- ✓ **Completion of the pilot plant : 2008**
- ✓ **Yield: 300 tons a year**
- ✓ **Method: melting and centrifugal casting**
- ✓ **Products: tube, plate, elbow, tee and other irregular parts**



The pilot plant of foam glass-ceramics

- ✓ **Completion of the pilot plant : 2011**
- ✓ **Capacity: 1000 cubic metres a year**
- ✓ **Method: sintering method**
- ✓ **Products: bearing brick, insulation board and anti-corrosion lining**

Products and Performance



Glass-ceramics tubes

The testing results from the
China Building Materials Academy

Item	Unit	result
Density	g/cm ³	3.21
Bending strength	MPa	192.25
Mohs hardness	1	9
Abrasion loss	g/cm ²	0.04
Impact toughness	kJ/m ²	2.06
Acid-resistance	%	98.35
Alkali-resistant	%	97.78

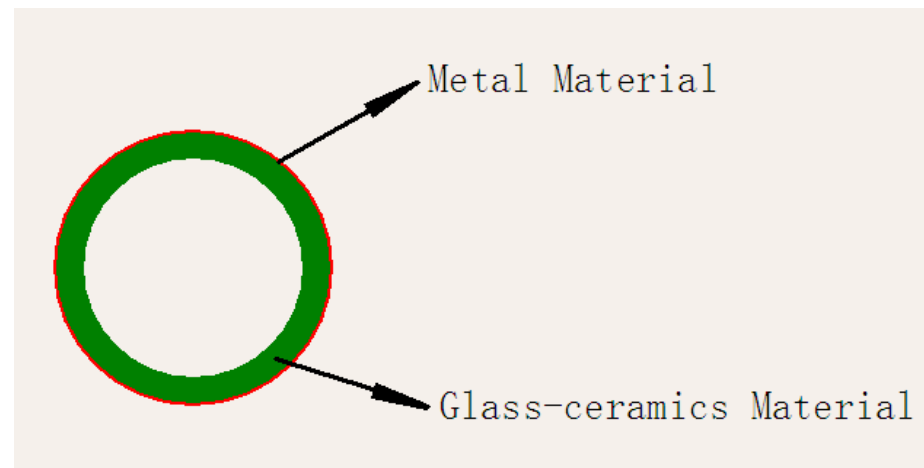
Products and Performance



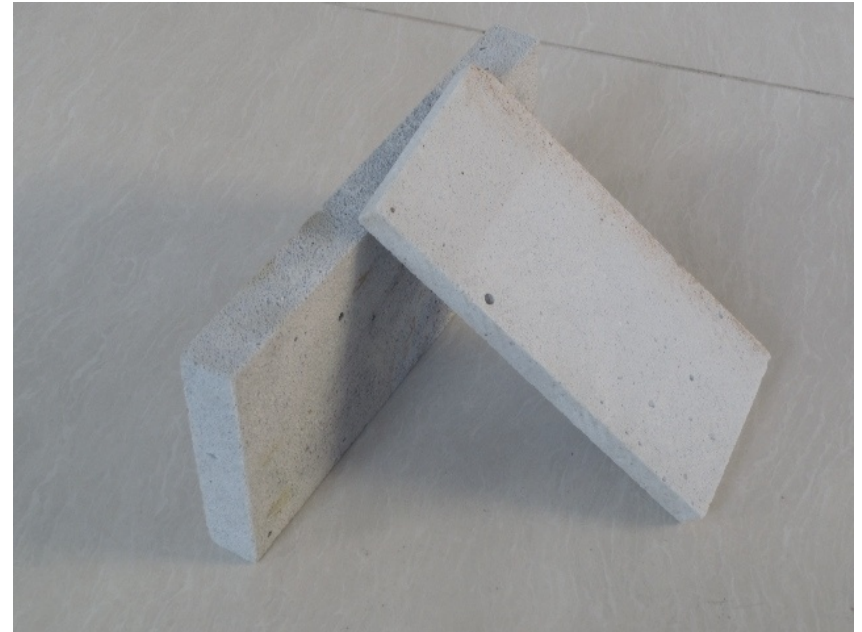
Glass-ceramics tubes

Products and Performance

Glass-
ceramics
compound
pipe
fitting



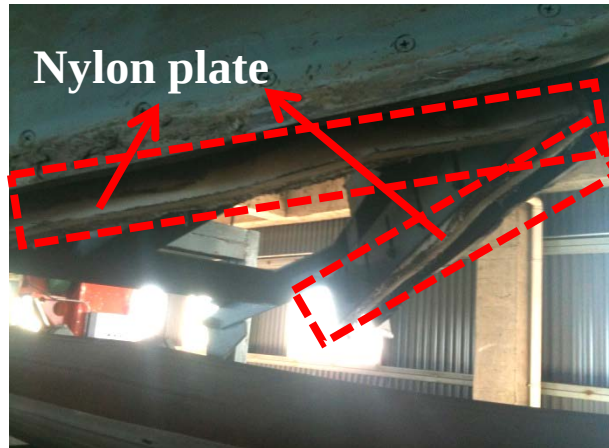
Products and Performance



Foam glass-ceramics bearing brick
and insulation board

Application and Promotion

❖ The application of slag glass-ceramics plate in metallurgical industry.



The service life of the nylon plate used previously was no more than 6 months. However, the service life of the slag glass-ceramics plate produced in our pilot plant was more than 3 years.

The application of slag glass-ceramics plate
in a metallurgical plant of Baotou Steel (Group) Co..

Application and Promotion



Slag glass-ceramics composite tube

❖ The application of slag glass-ceramics composite tube in cement industry.

The service life of the steel tube used previously was no more than **3 months**. However, the service life of the slag glass-ceramics composite tube produced in our pilot plant was more than **2 years**.

The application of slag glass-ceramics composite tube in a cement plant of Baotou Steel (Group) Co..

Application and Promotion

❖ The application of slag glass-ceramics composite tube in chemical industry.



The service life of the fiberglass tube used previously was no more than **2 months**. However, the service life of the slag glass-ceramics composite tube produced in our pilot plant was more than **1 year**.

The application of slag glass-ceramics composite tube in a chemical plant of Shenhua Group.

Application and Promotion

❖ The application of slag glass-ceramics composite elbow in power industry.



The service life of the wear-resistant metal elbow used previously was no more than **5 months**. However, the service life of the slag glass-ceramics composite elbow **produced in our pilot plant** was more than **2 years**.

The application of slag glass-ceramics composite elbow in a coal-fired power plant of Huaneng Group.

Innovative Pionts

- ❖ **Glass-ceramics with Excellent Structure and Performance**
 - ✓ abrasion resistant
 - ✓ corrosion resistant
 - ✓ high strength
 - ✓ nano-crystal
- ❖ **The Synergistic Mechanism of Trace and Valuable Elements**
 - ✓ glass phase separation and nucleation
 - ✓ refine grain
 - ✓ improve structure and performance
- ❖ **Good Formulas**
 - ✓ high-usage of solid waste
 - ✓ the wide range of compositions
- ❖ **Two Pilot Plants**
 - ✓ The slag glass-ceramics pilot plant
 - ✓ The foam glass-ceramics pilot plant



The Research Process, Achievement and Significance

Process

- **The First Stage: Basic Research (2000-2007)**
 - ✓ The fund resource: Self-financing and funds from the school
 - ✓ Research objectives: The development of glass ceramics formulation
- **The Second Stage: Pilot-Scale Study (2008-2013)**
 - ✓ The fund resource: The municipal and provincial projects
 - ✓ Research objectives: The development of glass ceramics formulation and the preparation process .
- **The Third Stage: Industrialization Precess(2014-)**
 - ✓ The fund resource: The provincial and National (973) projects
 - ✓ Research objectives: The development of the glass ceramics product, application and promotion



The Research Process, Achievement and Significance

Achievement

❖ Main Projects

- ✓ National Sci-Tech Support Plan : 2
- ✓ Major Project of Chinese National Programs for Fundamental Research and Development (973 Program):2
- ✓ National Natural Science Foundation: 16
- ✓ Inner Mongolia Science and Technology Innovation Guide Foundation: 5
- ✓ Inner Mongolia R&D Project : 4

The Research Process, Achievement and Significance

Achievement

❖ Representative papers

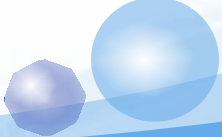
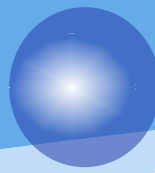
- [1] Li Baowei, Ouyang Shunli, Zhang Xuefeng. Effect of temperature on the structure of $\text{CaO-MgO-Al}_2\text{O}_3\text{-SiO}_2$ nanocrystalline glass-ceramics studied by Raman spectroscopy. *Spectroscopy and spectral analysis*. 2014, 34: 1869-1872.
- [2] Li Baowei, Deng Leibo, Zhang Xuefeng, Jia Xiaolin. Structure and performance of glass-ceramics obtained by Bayan Obo tailing and fly ash. *Journal of Non-crystalline Solids*. 2013, 380: 103-108.
- [3] Li Baowei, Du Yongsheng, Zhang Xuefeng et al. Effects of iron oxide on the crystallization kinetics of Baiyunebo tailing glass-ceramics. *Transactions of the Indian Ceramic Society*, 2013.
- [4] Li Baowei, Chen Hua, Zhao Ming, *et al.* Effect of CaO/SiO_2 Ratio and TiO_2 on Glass-Ceramics Made From Steel Slag and Fly Ash. *Applied Mechanics and Materials*, 2012, 217-219: 119-123.
- [5] Zhao Tuan, Wei Kaiming, Gao Zhanyong, *et al.* The Study on Crystallization Kinetics of Glass-ceramics Containing Rare-earth Tailings by DSC. *Advanced Materials Research*. 2011,199-120: 1741-1746.
- [6] Zhao Tuan, Li Baowei, Gao Zhanyong. The utilization of rare earth tailing for the production of glass-ceramics. *Materials Science and Engineering B*. 2010,170: 22-25.

The Research Process, Achievement and Significance

Achievement

❖ Patents

- [1] Rear-earth-rich, niobium-rich and cand-rich rear sorted tail-ore microcrystal glass and mfg. Method, CN200710084630.
- [2] Method for manufacturing glass-ceramics, CN200710195573.
- [3] Slag glass ceramic pipe and preparation method thereof, CN201110057464.
- [4] Compound tubular product of slag microcrystalline glass, CN201120061633.
- [5] Industrial wearable anti-corrosion glass ceramic composite bent pipe and manufacturing method thereof, CN201310133364.
- [6] Foam glass ceramic and preparation method thereof, CN201010537506.
- [7] Porous devitrified glass and manufacturing method thereof, CN200810144537.
- [8] Method for manufacturing low-swelling microcrystalline glass through microwave heat treatment of gold Tailings, CN201310067629.
- [9] Method for preparing light wall material by microwave baking of chromic slag, CN201210589168.
- [10] Double-tank type electric melting furnace, CN201310158180.
- [11] Electric melting furnace for testing, CN201320231863.



The Research Process, Achievement and Significance

Achievement

❖ Industry standard

JC/T 2283-2014 The slag glass-ceramics pipes.

The top of the slide features a blue wavy banner. On the left, there is a large blue sphere. On the right, there are two smaller blue spheres, one of which is partially obscured by the edge of the frame.

The Research Process, Achievement and Significance

Significance

- ❖ The research and development of the slag glass-ceramics material prepared from the waste from mineral resources which is extremely important to the comprehensive utilization of solid waste.
- ❖ Largely depletion of waste will achieve the goal of the secondary resource recycle.
- ❖ The research and development of the slag glass-ceramics establishes a foundation for solving the stubborn problems, such as the land waste, water and environmental pollution, etc.



Thank you!